SPACE MISSION LAUNCH ANALYSIS

1. INTRODUCTION

Space exploration has changed over the years with many organisations launching space missions from various parts of the world. In this project we analyse the launch trends, mission outcomes, organisational activities and recorded launch costs.

This project uses Python (pandas, matplotlib) to explore and identify the pattern and trends in space mission launch dataset

The dataset was downloaded from kaggle, an open source website which you can find here.

2. PROJECT OBJECTIVES

The main objectives of the project are:

- Identify the organisations that conducted the most launches.
- Compare successful and unsuccessful mission outcomes.
- Analyse how launch activity changed over time.
- Identify the most frequently used launch locations.
- Compare average recorded launch costs across organisations.
- Examine success rates for organisations with at least 10 recorded launches.

3. TOOLS AND TECHNOLOGIES

- Python - Programming language used
- Pandas - Used for data cleaning, analysis and manipulation.
- Matplotlib - Used for data visualisation.
- Jupyter Notebook - environment used to conduct and document the analysis.

4. DATASET OVERVIEW

The dataset consists of total 4,324 records of space missions and information such as :

- Organisation
- Launch location
- Launch date
- Launch detail
- Rocket status
- Recorded launch price
- Mission status

5. EXPLORATORY DATA ANALYSIS

5.1 Determining which organisation conducted the most mission launches.

When the organisations were grouped together to identify which organizations conducted the most launches it was found that *RVSN USSR* had the highest number of mission launches. However, having the highest space mission doesn't automatically mean the highest success rate.

5.2 Determining how many launches were successful vs unsuccessful

Mission status was distributed in 4 groups: successful, Failure, Partial Failure, Prelaunch Failure. It was found that there were 3879 successful launches and 339 failures whereas there were 102 partial failures and 4 pre-launch failures. These are total successful launches including all the organization.

5.3 Identifying how number of launches changed over the years

It was noticed that mission launches fluctuated over the years but showed significant peaks 1967-1971 , 1975-1978 and 2018-2019. It is important to note that 2020 data ends in August so it represents an incomplete year.

5.4 Organisation that have the highest number of successful missions

RVSN USSR has the highest successful launches. It is important to note that the highest number of successful launches does not mean the highest success rates.

5.5 Most Frequently Used Launch Location

The analysis showed that Site 31/6, Baikonur Cosmodrome, Kazakhstan, was the most frequently used launch location, recording 235 launches. This highlights the importance of this specific launch site in the space mission records included in the dataset.

5.6 Organisational Success Rates

The success rates of organisations with at least 10 recorded launches were compared to make the results more meaningful and reduce the influence of very small sample sizes. This analysis provides a different perspective from comparing the total number of successful missions, as it measures the proportion of recorded missions that were successful.

5.7 Average Launch Cost

The average recorded price was calculated for the dataset and separately for each organisation. The overall average recorded launch cost was approximately 129.80. NASA had the highest average among the listed organisations at 450.00, followed by Boeing at 177.29 and Arianespace at 170.26.

These figures should be interpreted cautiously because many missions do not have recorded prices and the dataset's price units need confirmation.

5.8 Launch Cost by Mission Outcome

Average recorded prices were calculated for successful and failed missions to examine whether costs differed between the two groups. The comparison uses only missions with recorded prices.

Failure 67.359444

Success 133.316451

The final conclusion should be based on the actual calculated averages.

6. LIMITATIONS

- Price information is missing for many missions.
- The dataset's price units need to be confirmed.
- The 2020 records cover only part of the year.
- Historical launch counts and success rates describe the available dataset and do not guarantee future performance.
- Differences in recorded launch costs do not establish that cost causes mission success or failure.

7. CONCLUSION

This project used Python, Pandas, and Matplotlib to explore space mission launch dataset. The analysis examined launch activity by organisation, mission outcomes, yearly trends, launch locations, recorded costs, and organisational success rates.

The project demonstrates how data cleaning, aggregation, and visualisation can be used to turn raw data into meaningful insights. It also highlights the importance of considering missing values, incomplete time periods, and the difference between total successful missions and success rates when interpreting analytical results.